1 Gain law

(bars denote normalized quantities: lengths by R, $\bar a = a/a_o$, $\bar f = a_o f$; $0 < \bar a < 1$; $e_\theta = \theta - \hat\theta$)

$$\bar a_{\text{true}}(\bar w) = 1 - \frac{9}{8\bar w} + \frac{1}{2\bar w^3} - \frac{57}{100\,\bar w^4} + \frac{1}{5\bar w^5} + \frac{7}{200\,\bar w^{11}} - \frac{1}{25\,\bar w^{12}}$$

$$\bar a'_{\text{true}}(\bar w) > 0 \ \text{ on } \ \bar w > 1 \qquad \Longrightarrow \qquad \bar a' = b_{\text{true}}(\bar a)\big(1-\bar a\big)^2 \ \text{ exactly}, \qquad b_{\text{true}} := \frac{\bar a'_{\text{true}}}{\big(1-\bar a_{\text{true}}\big)^2}$$

$$\begin{aligned} \bar a_{\text{true}}(1) = 0, \quad \bar a'_{\text{true}}(1) = 1 \quad &\Longrightarrow \quad b_{\text{true}}\big(\bar a = 0\big) = 1 \\ 1 - \bar a_{\text{true}} = \frac{9}{8\bar w} + \mathcal{O}(\bar w^{-3}) \quad &\Longrightarrow \quad b_{\text{true}}\big(\bar a \to 1\big) = \frac{8}{9} \end{aligned}$$

$$\bar a'(\bar a, b) := b\big(1-\bar a\big)^2$$
$$a_w = a_o\,\bar a_w, \qquad a_o = \frac{\Delta t}{\gamma_N R}$$

2 Jacobians

$$\frac{\partial \bar a'}{\partial \bar a} = -2b\big(1-\bar a\big)$$
$$\frac{\partial \bar a'}{\partial b} = \big(1-\bar a\big)^2 = \frac{\bar a'}{b}$$

3 True dynamics

$\big(\delta\bar w_j[k] := \delta\bar w[k-d+j-1],\ j = 1,\dots,d+1,\ d = 2\big)$

$$\begin{aligned} \bar w[k] &= \bar w_d[k] - \delta\bar w_3[k] \\ \bar w_d[k+1] &= \bar w_d[k] + \Delta\bar w_d[k] \\ \delta\bar w_1[k+1] &= \delta\bar w_2[k] \\ \delta\bar w_2[k+1] &= \delta\bar w_3[k] \\ \delta\bar w_3[k+1] &= \lambda_c\,\delta\bar w_3[k] - F_{dw}[k]\,e_{\bar a_w}[k] - \varepsilon_w[k] \\ \bar a_w[k+1] &= \bar a_w[k] + \bar a'_{\text{true}}[k]\Big[\Delta\bar w_d[k] + (1-\lambda_c)\,\delta\bar w_3[k] + F_{dw}[k]\,e_{\bar a_w}[k] + \varepsilon_w[k]\Big] \\ b[k+1] &= b[k] \end{aligned}$$

$$F_{dw}[k] = \bar{f}_{dw}[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{f}_{dw}[k-i]$$

$$\varepsilon_w[k] = \bar{a}_w[k]\,\bar{f}_T[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{a}_w[k-i]\,\bar{f}_T[k-i] + (1-\lambda_c)\,n_w[k]$$

4 Process model

$$\mathbf{x} = \left[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ b\,\right]^\top \qquad (d=2)$$

5 Estimator

$$\begin{aligned}
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k]\\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k]\\
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k]\\
\hat{\bar{a}}_w[k+1] &= \hat{\bar{a}}_w[k] + \hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)^2\Big[\Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]\Big]\\
\hat{b}[k+1] &= \hat{b}[k]
\end{aligned}$$

6 Error dynamics

(True − Estimator; coefficients at the estimate)

$$\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k]\\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k]\\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c\,e_{\delta\bar{w}_3}[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k]\\
e_{\bar{a}_w}[k+1] &= \Big[\,1 + \hat{b}\big(1-\hat{\bar{a}}_w\big)^2 F_{dw} - 2\hat{b}\big(1-\hat{\bar{a}}_w\big)\big(\Delta\bar{w}_d + (1-\lambda_c)\delta\hat{\bar{w}}_3\big)\Big]e_{\bar{a}_w}[k]\\
&\quad + (1-\lambda_c)\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 e_{\delta\bar{w}_3}[k] + \big(1-\hat{\bar{a}}_w\big)^2\big(\Delta\bar{w}_d + (1-\lambda_c)\delta\hat{\bar{w}}_3\big)e_b[k]\\
&\quad + \hat{b}\big(1-\hat{\bar{a}}_w\big)^2\varepsilon_w[k]\\
e_b[k+1] &= e_b[k]
\end{aligned}$$

$$F_e[k] = \begin{bmatrix}
0 & 1 & 0 & 0 & 0\\
0 & 0 & 1 & 0 & 0\\
0 & 0 & \lambda_c & -F_{dw} & 0\\
0 & 0 & (1-\lambda_c)\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 & \begin{array}{c}1 + \hat{b}\big(1-\hat{\bar{a}}_w\big)^2 F_{dw}\\ -2\hat{b}\big(1-\hat{\bar{a}}_w\big)\big(\Delta\bar{w}_d + (1-\lambda_c)\,\delta\hat{\bar{w}}_3\big)\end{array} & \begin{array}{c}\big(1-\hat{\bar{a}}_w\big)^2\\ \times\big(\Delta\bar{w}_d + (1-\lambda_c)\,\delta\hat{\bar{w}}_3\big)\end{array}\\
0 & 0 & 0 & 0 & 1
\end{bmatrix}$$

$$\mathbf{q}[k] = \Big[0,\ 0,\ -\varepsilon_w,\ \hat{b}\big(1-\hat{\bar{a}}_w\big)^2\varepsilon_w,\ 0\Big]^\top$$

7 Measurement and innovation

$$\begin{aligned}
\nabla_d \bar{w}_d[k] &= \bar{w}_d[k] - \bar{w}_d[k-d] \\
y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_w[k-d] + n_a[k]
\end{aligned}$$

$$\bar{a}_w[k-d] = \bar{a}_w[k] - b\big(1-\bar{a}_w[k]\big)^2\nabla_d\bar{w}_d[k]$$

$$\begin{aligned}
\frac{\partial y_2}{\partial \bar{a}_w} &= 1 + 2\hat{b}\big(1-\hat{\bar{a}}_w\big)\nabla_d\bar{w}_d \\
\frac{\partial y_2}{\partial b} &= -\big(1-\hat{\bar{a}}_w\big)^2\nabla_d\bar{w}_d
\end{aligned}$$

$$H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1+2\hat{b}(1-\hat{\bar{a}}_w)\nabla_d\bar{w}_d & -(1-\hat{\bar{a}}_w)^2\nabla_d\bar{w}_d \end{bmatrix}$$

$$L[k] = P^f H^\top\big(HP^f H^\top + R\big)^{-1}$$

$$\begin{aligned}
e_{y_1}[k] &= e_{\delta\bar{w}_1}[k] + n_w[k] \\
e_{y_2}[k] &= \Big[1+2\hat{b}\big(1-\hat{\bar{a}}_w\big)\nabla_d\bar{w}_d\Big]e_{\bar{a}_w}[k] - \big(1-\hat{\bar{a}}_w\big)^2\nabla_d\bar{w}_d\ e_b[k] + n_a[k]
\end{aligned}$$

8 Process and measurement noise

$$\begin{aligned}
q_3 &= -\bar{a}_w\,\bar{f}_T \\
q_4 &= -\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 q_3 \\
\gamma(\bar{w}) &= \gamma_N c(\bar{w}) \\
\sigma^2_{\bar{f}_T} &= a_o^2\,\frac{4k_B T\gamma(\bar{w})}{\Delta t}
\end{aligned}$$

$$\begin{aligned} Q_{33} &= \frac{4k_B T\, a_o}{R}\, \bar{a}_w \\ Q_{34} = Q_{43} &= -\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 Q_{33} \\ Q_{44} &= \hat{b}^2\big(1-\hat{\bar{a}}_w\big)^4 Q_{33} \\ Q_{55} &= 0 \end{aligned}$$

$$\begin{aligned} R &= \mathrm{diag}(R_1, R_2) \\ R_1 &= \sigma_{n_w}^2 \\ R_2 &= K_{\mathrm{var}} IF_{\mathrm{var}}\big(\bar{a}_w + \xi\big)^2 + d\, Q_{44} \\ K_{\mathrm{var}} &= \frac{2a_{\mathrm{cov}}}{2-a_{\mathrm{cov}}} \\ \sigma_{\delta\bar{w}_r}^2 &= C_{dpmr}\frac{4k_B T\, a_o}{R}\big(\bar{a}_w + \xi\big) \\ \xi &= \frac{C_n \sigma_{n_w}^2}{C_{dpmr}\big(4k_B T a_o/R\big)} \end{aligned}$$

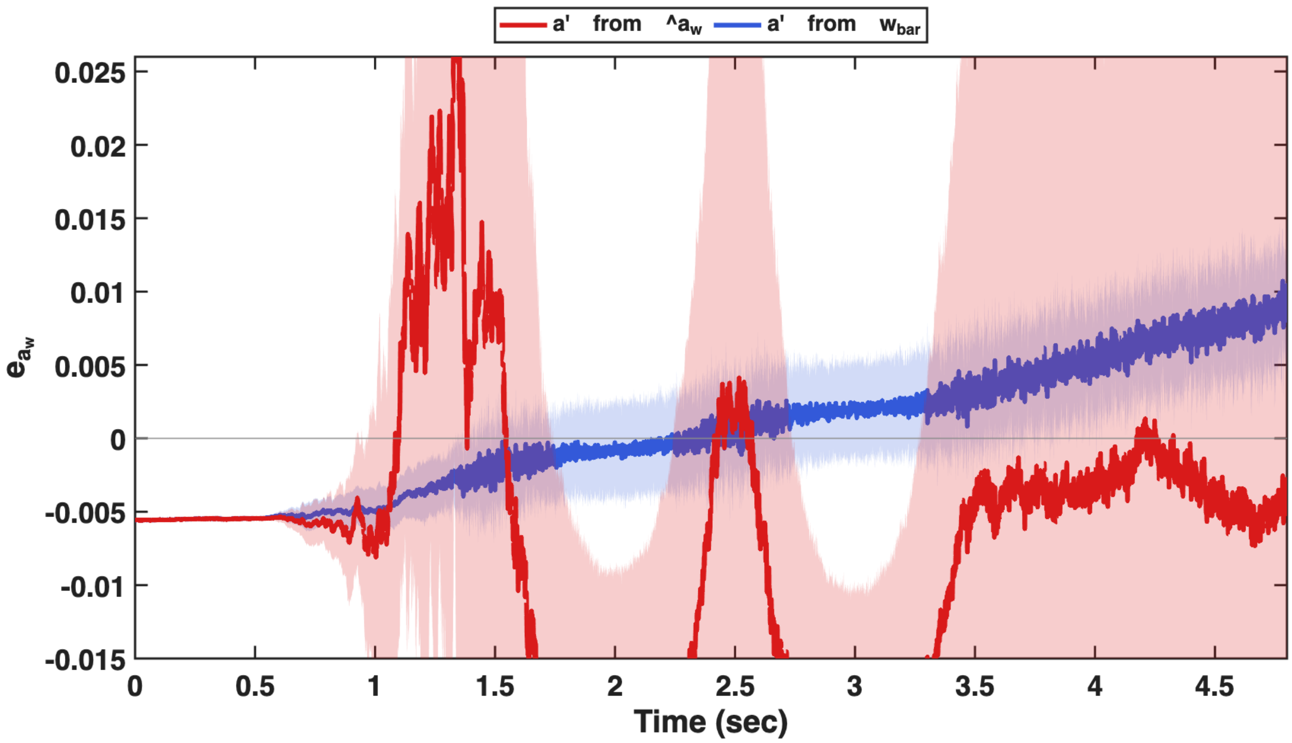
Seed and prior

$$\begin{aligned} \bar{a}(\bar{w}) &= 1 - \frac{1}{b\,(\bar{w}-\bar{w}_s)} \\ \left.\frac{\partial \bar{a}}{\partial b}\right|_{\bar{w}} &= \frac{1}{b^2(\bar{w}-\bar{w}_s)} = \frac{1-\bar{a}}{b} \\ \left.\frac{\partial \bar{a}}{\partial \bar{w}_s}\right|_{\bar{w}} &= -\,\bar{a}' \end{aligned}$$

$$\begin{aligned} \hat{b}[0] &= \tfrac{1}{2}\Big[\inf_{\bar{w}\in\mathcal{E}} b_{\mathrm{true}} + \sup_{\bar{w}\in\mathcal{E}} b_{\mathrm{true}}\Big] \\ P_{55}[0] = P_{bb} &= \Big[\tfrac{1}{2}\Big(\sup_{\bar{w}\in\mathcal{E}} b_{\mathrm{true}} - \inf_{\bar{w}\in\mathcal{E}} b_{\mathrm{true}}\Big)\Big]^2 \\ \hat{\bar{a}}_w[0] &= 1 - \frac{1}{\hat{b}[0]\,(\bar{w}[0]-\hat{\bar{w}}_s)} \end{aligned}$$

$$\begin{aligned} P_{44}[0] &= \big(\hat{\bar{a}}'[0]\big)^2 P_{\bar{w}_s} + \left(\frac{1-\hat{\bar{a}}_w[0]}{\hat{b}[0]}\right)^2 P_{bb} + \mathrm{floor}_a^2 \\ P_{45}[0] &= \frac{1-\hat{\bar{a}}_w[0]}{\hat{b}[0]}\, P_{bb} \end{aligned}$$

$$\begin{aligned} e_{\bar{a}}(\bar{w})\big|_{\bar{w}_s} &= -\,\bar{a}'(\bar{w})\, e_{\bar{w}_s} \;\propto\; \frac{1}{b\,(\bar{w}-\bar{w}_s)^2} \\ e_{\bar{a}}(\bar{w})\big|_{b} &= \frac{e_b}{b^2(\bar{w}-\bar{w}_s)} \end{aligned}$$


a' from ^aw
a' from wbar
0.025
0.02
0.015
0.01
0.005
0
-0.005
-0.01
-0.015
e_aw
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
Time (sec)